

Syllabus:- Statistics and Mathematical Reasoning

SECTION-I
Straight Objective Type

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- The average marks of 10 students in a class is 60 with standard deviation 4 while the average marks of other 10 students is 40 with standard deviation 6. If all the 20 students are taken together, their standard deviation will be
 (1) $\sqrt{109}$ (2) 12 (3) $\sqrt{116}$ (4) $\sqrt{126}$
- n observations on a variable X_i are $X_i = A + iB$ for $i = 1, 2, 3, \dots, n$ where A, B are real constants. The mean of these observation is
 (1) $A + B \left(\frac{n+1}{2} \right)$ (2) $nA + B \left(\frac{n+1}{2} \right)$ (3) $A + Bn \left(\frac{n+1}{2} \right)$ (4) $A + B \left(\frac{n}{2} \right)$
- Coefficient of variance of a distribution is 60% and the standard deviation is 25. The arithmetic mean of the distribution is
 (1) $\frac{25}{3}$ (2) 35 (3) $\frac{125}{3}$ (4) $\frac{25}{6}$
- The mean deviation from mean of observations $a, a + d, a + 2d, \dots, a + 2nd$ is
 (1) $\frac{n(n+1)d^2}{3}$ (2) $\frac{n(n+1)d^2}{2}$ (3) $a + nd$ (4) $\frac{n(n+1)}{2n+1} |d|$
- In a set of $2n$ distinct observations, each of the observation below the median of all the observations is increased by 5 and each of the remaining observations is decreased by 3. Then the mean of the new set of observations :
 (1) increases by 1 (2) decreases by 1 (3) decreases by 2 (4) increases by 2
- The mean and the median of 100 observations have been computed to be 60 and 70 respectively. Later it was discovered that three observations which have been recorded as 18, 28 and 98 are actually 80, 26 and 38 respectively. If the mean and median are recalculated with actual observations, then :
 (1) median will change but mean will not change
 (2) neither mean nor median will change
 (3) both mean and median will change
 (4) mean will change but median will not change

7. The weighted mean of first n natural numbers whose weights are equal to the squares of the corresponding numbers is
- (1) $\frac{n+1}{2}$ (2) $\frac{3n(n+1)}{2(2n+1)}$ (3) $\frac{(n+1)(2n+1)}{6}$ (4) $\frac{n(n+1)}{2}$
8. The mean marks of 120 students of a school is 56. The mean marks of girls is 60 and that of boys is 50. If number of boys and girls are n and m respectively, then $\frac{9n}{m}$ equals.
- (1) 2 (2) 4 (3) 6 (4) 8
9. The mean & variance of five observations are 4 & 5.20, respectively. If three of the observations are 3, 4, and 4; then the absolute value of difference of other two observations, is
- (1) 3 (2) 5 (3) 7 (4) 9
10. The sum of deviations of 50 observations from 30 (it means sum of (variate -30)) is 50. Then sum of digits in the mean of these observations is:
- (1) 1 (2) 2 (3) 3 (4) 4
11. A data has highest value 120 and lowest value 71. A frequency distribution in descending order with seven classes is to be constructed. The limits of the second class–interval shall be
- (1) 71 and 78 (2) 78 and 85 (3) 113 and 120 (4) 106 and 113
12. The mean of n items is $\bar{x}$. If the first term is incremented by 1, second term by 2 and so on, then the new mean is
- (1) $\bar{x} + \frac{n+1}{2}$ (2) $\bar{x} - \frac{n+1}{2}$ (3) $\bar{x} + \frac{n}{2}$ (4) $\bar{x} + n$
13. The first of the two samples in a group has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{13.44}$, then the standard deviation of the second sample is :
- (1) 8 (2) 6 (3) 4 (4) 5
14. If the mean deviation about the mean of the numbers 1, 2, 3,, n , where n is odd, is $\frac{5(n+1)}{n}$, then n is equal to _____.
- (1) 20 (2) 10 (3) 12 (4) 21
15. The mean of the numbers a , b , 8, 5, 10 is 6 and their variance is 6.8. If M is the mean deviation of the numbers about the mean, then $25M$ is equal to :
- (1) 60 (2) 55 (3) 50 (4) 45

16. The mean and standard deviation of 50 observations are 15 and 2 respectively. It was found that one incorrect observation was taken such that the sum of correct and incorrect observations is 70. If the correct mean is 16, then the correct variance is equal to :
 (1) 10 (2) 36 (3) 43 (4) 60
17. Let a Relation R on set N of natural numbers is defined as $(x,y) \Rightarrow x^2 - 4xy + 3y^2 = 0 \forall x,y \in N$. Relation R is
 (1) reflexive (2) symmetric (3) transitive (4) equivalence
18. The mean and variance of the data 4, 5, 6, 6, 7, 8, x, y where $x < y$ are 6, and $\frac{9}{4}$ respectively. Then $x^4 + y^2$ is equal to
 (1) 162 (2) 320 (3) 674 (4) 420
19. The mean and standard deviation of 15 observations are found to be 8 and 3 respectively. On rechecking it was found that, in the observations, 20 was misread as 5. Then, the correct variance is equal to _____.
 (1) 16 (2) 32 (3) 67 (4) 17
20. Suppose a class has 7 students. The average marks of these students in the mathematics examination is 62, and their variance is 20. A student fails in the examination if he/she gets less than 50 marks, then in worst case, the number of students can fail is
 (1) 1 (2) -1 (3) 2 (4) 0

SECTION-II

(Integer Type Questions)

This section contains **Five (5)** questions. The answer to each question is **NUMERICAL VALUE** with two digit integer and decimal upto one digit

21. If the standard deviation of $x_1, x_2, \dots, x_n$ is 3.5 then the standard deviation of $-2x_1-3, -2x_2-3, \dots, -2x_n-3$ is
22. The mean deviation about mean of an ungrouped data is 10. If each observation is increased by 4%, the revised mean deviation about new mean is
23. In a class of 20 students, each student can score either 10 or 0 marks in a certain examination. The maximum possible variance in the marks of the students in the class is

24. If the mean and the standard deviation of 10 observations $x_1, x_2, \dots, x_{10}$ are 2 and 3 respectively, then the mean of $(x_1 + 1)^2, (x_2 + 1)^2, \dots, (x_{10} + 1)^2$ is equal to :
25. In a frequency distribution, the mean and median are 21 and 22 respectively. Its mode is

ANSWER_KEY_DPP-21

1.	(4)	2.	(1)	3.	(3)	4.	(4)	5.	(1)	6.	(2)	7.	(2)
8.	(3)	9.	(3)	10.	(4)	11.	(4)	12.	(1)	13.	(3)	14.	(4)
15.	(1)	16.	(3)	17.	(1)	18.	(2)	19.	(4)	20.	(4)	21.	(7)
22.	(10.4)	23.	(25)	24.	(18)	25.	(24)						